EFFECTIVE NUMERICAL COMPUTATION OF $p(x)-$LAPLACE EQUATIONS IN 2D

ADRIANA ARAGÓN, JULIÁN FERNÁNDEZ BONDER AND DIANA RUBIO

Abstract. In this article we implement a method for the computation of a nonlinear elliptic problem with nonstandard growth driven by the $p(x)-$Laplacian operator. Our implementation is based in the *decomposition–coordination* method that allows us, via an iterative process, to solve in each step a linear differential equation and a nonlinear algebraic equation. Our code is implemented in MatLab in 2 dimensions and turns out to be extremely efficient from the computational point of view.

1. Introduction

In this article we consider nonlinear elliptic problems with nonstandard growth. More precisely, we will focus on the so-called $p(x)-$Laplace operator that is defined as

$$\Delta_{p(x)}u = \nabla \cdot \left(|\nabla u|^{p(x)-2}\nabla u \right).$$

This operator has been used by several authors in view of its applications to several fields of sciences such as electrorheological fluids and image processing. We may cite the works [4, 8] and references therein.

From the mathematical point of view, this operator presents interesting difficulties since when the exponent $p(x)$ is not a constant, the operator is no longer homogeneous.

Recall that when $p(x) = p$ constant the $p(x)-$Laplacian becomes the well-known $p-$Laplacian operator that is well studied in the literature both from the theoretical point of view and also from the numerical analysis. See [1], for instance.

In this work we present a numerical implementation of the so-called *decomposition-coordination method* to solve $p(x)-$Laplacian problems of the form

$$\begin{cases} -\Delta_{p(x)}u = f & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega, \end{cases} \tag{1.1}$$

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where Ω is a Lipschitz domain in $\mathbb{R}^2$ and $f\colon \Omega \to \mathbb{R}, g\colon \partial\Omega \to \mathbb{R}$ are suitable data.

The implementation is done in MatLab but is an arbitrary choise and our code can be easily adapted to any other programming language.

The problem of efficiently computing a solution to (1.1) presents several difficulties due to the strong nonlinearity and more importantly due to the degenerate/singular character of the operators in points $x \in \Omega$ where $\nabla u(x) = 0$ depending on $p(x) > 2$ or $p(x) < 2$.

As far as we know, the first rigorous analysis for numerical approximations for (1.1) was done in [5]. In that article the authors studied the approximation of (1.1) by means of the Finite Element Method and proved the convergence of the method, See also [2, 7].

In a subsequent paper, [6], the authors introduce the so-called *decomposition-coordination method* for problem (1.1) that is an iterative procedure that has the advantage that in each step of the iteration one has to solve a *linear PDE* and a *nonlinear algebraic equation* which in theory makes the problem much more tractable and efficient from a computational point of view.

The purpose of this work is to implement a simple and efficiently the method developed in [6] that allow to solve (1.1) in dimension 2.

To end this introduction, let us mention the work [3] where the authors numerically solve (1.1) with very different techniques that the ones presented here.

2. Description of the method

In this section we give a description of the decomposition-coordination method.

This method is an iterative procedure that has the following form: First one define two vector fields $\eta_1, \nu_0\colon \Omega \to \mathbb{R}^2$.

Then, recursively, if we have computed u_{n-1}, η_n and ν_{n-1} we compute u_n, η_{n+1} and ν_n in the following form:

(1) Solve the *linear PDE*

$$-\Delta u_n = \nabla \cdot (\eta_n - \nu_{n-1}) + f \tag{2.1}$$

(2) Update ν by solving the *algebraic nonlinear equation*

$$|\nu_n|^{p-2}\nu_n + \nu_n = \eta_n + \nabla u_n. \tag{2.2}$$

(3) Finally, update η as

$$\eta_{n+1} = \eta_n + \nabla u_n - \nu_n. \tag{2.3}$$

In [6] it is proved that this procedure is in fact convergent and that the sequence $\{u_n\}_{n\in\mathbb{N}}$ defined by (2.1) converges to the actual solution u of (1.1).

Have in mind that $p = p(x)$ in (2.2).

Observe that in this method, in each step of the iteration, one must sove a linear PDE. This PDE is discretized using the finite element method. The matrix of the method is the same in each step.

Also observe that the nonlinear algebraic equation (2.2) that is a vectorial equation, can be easily solved in this form: denote $t_n = |\nu_n|$, and then $t \in \mathbb{R}$ is the solution to

$$t_n(t_n^{p-2} + 1) = r_n,$$

where $r_n = |\eta_n + \nabla u_n|$. This can be easily solved using, for instance, the bisection method or Newton-Raphson method. Once that t_n is computed, one solve

$$\nu_n = \frac{\eta_n + \nabla u_n}{t_n^{p-1} + 1}.$$

Step 3 in the iteration is a simple evaluation.

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3. Numerical experiments

In this section we illustrate our implementation by comparing some exact solutions to the ones computed by our method.

To begin with let us consider $p(x) = p$ constant and look for two cases, one with $p > 2$ where the equation (1.1) is degenerate and other with $1 < p < 2$ in which equation (1.1) is singular. Then we give an example with variable exponent $p(x)$.

In all of the cases we give a preliminar error analysis.

Example 1. In this first example, we consider a constant exponent $p > 2$. In this case, the function

$$u(x, y) = (x^2 + y^2)^{\frac{p-2}{2p-2}}$$

verifies $-\Delta_p u = 0$.

We take $\Omega = (0, 1) \times (0, 1)$, $p = 20$ and as a boundary data the same u.

We make a regular partition of Ω of 100×100.

If we denote by u_h the computed solution, the errors obtained are

$$\|u - u_h\|_\infty \sim 2 \times 10^{-3}, \qquad \|u - u_h\|_p \sim 10^{-3}.$$

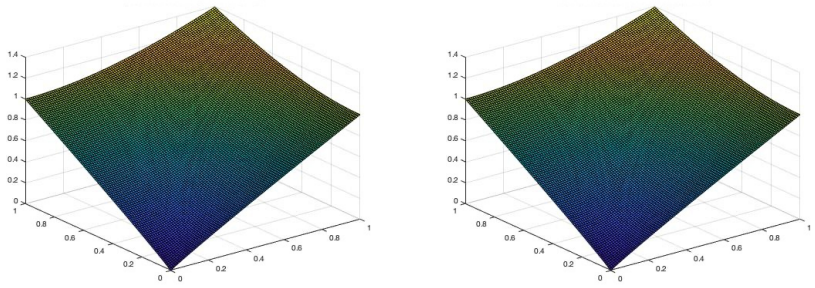


Figure 1. The exact solution at the left and the computed solution at the right

The time employed in the iteration was $\sim 75s$.

Example 2. In this example, we consider the so-called *torsion problem*,

$$-\Delta_p u = 1.$$

An exact solution to this problem is given by

$$u(x,y) = c_p \left(1 - (x^2+y^2)^{\frac{p}{2p-2}}\right),$$

where c_p is a constant, $c_p = \frac{(p-1)}{p} 2^{-1/(p-1)}$.

Let us consider the domain $\Omega = (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \times (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and constant p, $p = 1.1$. As boundary data we consider the same u and again we make a regular partition of the domain Ω of 100×100.

As in the previous example, we call u_h the computed solution and the errors obtained are

$$\|u - u_h\|_\infty \sim 2 \times 10^{-4}, \qquad \|u - u_h\|_p \sim 3 \times 10^{-4}.$$

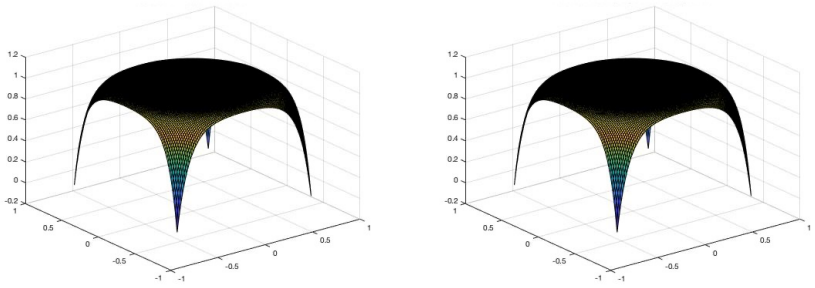


FIGURE 2. The exact solution at the left and the computed solution at the right

The time employed in the iteration was $\sim 80s$.

Example 3. In this example we consider a variable exponent p

$$p(x,y) = 1 + \left(\frac{x+y}{2} + 2\right)^{-1} \quad \text{and} \quad u(x,y) = \sqrt{2}e^2(e^{\frac{1}{2}(x+y)} - 1).$$

This function u is a solution to $-\Delta_{p(x,y)} u = 0$.

Consider now the domain $\Omega = (-1,1) \times (-1,1)$ with a regular partition of 100×100 and take as boundary data the same u.

Again, calling u_h to the computed solution, the errors obtained are

$$\|u - u_h\|_\infty \sim 1.5 \times 10^{-5}, \qquad \|u - u_h\|_p \sim 2.7 \times 10^{-4}.$$

The time employed in the iteration was $\sim 81s$.

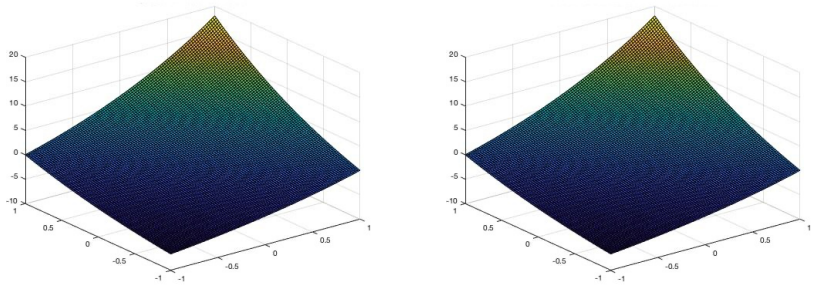


FIGURE 3. The exact solution at the left and the computed solution at the right

Example 4 - The torsion problem. In this example, we work with the *torsion problem*, with homogeneous Dirichlet boundary data and constant exponent p. This is

$$\begin{cases} -\Delta_p u = 1 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

An exact solution to this problem is not available. We consider $\Omega = (-1,1)\times(-1,1)$ and a regular partition of 100×100. We obtain the following computed solutions for different values of the exponent p.

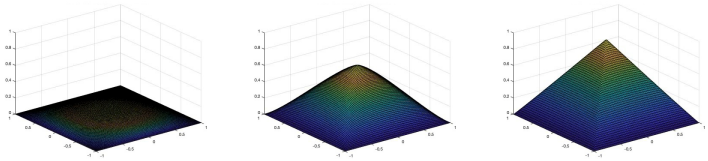


FIGURE 4. To the left, the computed solution with $p = 1.15$, in the center the computed solution with $p = 4$ and to the right the computed solution with $p = 50$

Let us observe that for small values of p the solution is *flat* and as the value of p increases, the solution is getting closer to the *pyramid* with maximum at the center of the domain.

Example 5 - variable exponent. Finally, we analyze the torsion problem for variable exponents

$$\begin{cases} -\Delta_{p(x)} u = 1 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

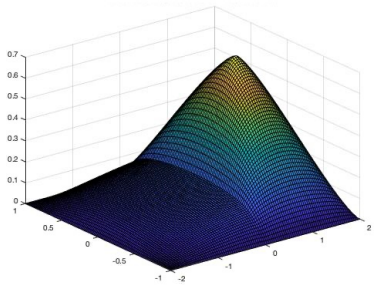


FIGURE 5. The torsion problem with variable and discontinuous exponent

We consider a *discontiuous* exponent,

$$p(x,y) = \begin{cases} 1.2 & x \leq 0, \\ 4 & x > 0 \end{cases}$$

and a rectangular domain $\Omega = [-2,2] \times [-1,1]$.

Again, an exact solution is not available.

Our method can handle also this case and the computed solution is shown in Figure 5

As expected, the solution is *flat* in the region $\{x \leq 0\}$ where the exponent is small ($p(x,y) = 1.2$) and has a *pyramid shape* in the region $\{x > 0\}$ where the exponent is large ($p(x,y) = 4$).

The time employed in the iteration was $\sim 52s$.

4. Conclusions

We implemented a Finite Element based algorithm for the computation of solutions of equations with nonstandard growth of $p(x)-$Laplacian type. This algorithm uses the decomposition-coordination method that has the advantage that it is an iterative scheme that in each step of the iteration one has to solve a linear PDE and a nonlinear algebraic equation. Our numerical examples show that this method is extremely efficient for these cases.

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